1. Readiness and the SBT

Readiness

- Cause improving, P/F > 150–200 on PEEP ≤ 8, FiO2 ≤ 0.4–0.5
- Haemodynamically stable (little or no vasopressor), awake (RASS 0/−1), cough, manageable secretions

SBT

- 30–120 min T-piece or PSV 5–8 / PEEP 0–5
- Failure: RR > 35, SpO2 < 90%, HR > 140 or ↑ 20%, SBP > 180 or < 90, agitation, diaphoresis
- RSBI (f/Vt) < 105 supports success

Classification (Boles 2007 / WIND)

- Simple (first SBT), difficult (≤ 3 SBTs or ≤ 7 days), prolonged (> 7 days)

Key point: Daily screen + daily SBT + paired sedation hold (ABC trial).

2. Causes of weaning failure (ABCDE)

Load vs capacity

- **A**irway/lung: secretions, bronchospasm, ↑ resistance, PEEPi (COPD), effusion, oedema
- **B**rain: delirium, sedation, anxiety/psychological dependence
- **C**ardiac: weaning-induced pulmonary oedema (ischaemia, LV dysfunction)
- **D**iaphragm: ventilator-induced diaphragm dysfunction, phrenic injury, ICU-AW, electrolytes (↓ PO4, K, Mg), hypothyroid
- **E**ndocrine/metabolic: metabolic alkalosis, nutrition, anaemia

Key point: Frame it as respiratory load versus neuromuscular capacity, plus cardiac and psychological.

3. Weaning-induced pulmonary oedema

Mechanism

- Negative pleural pressure ↑ venous return and LV afterload; catecholamine surge; ischaemia

Diagnosis

- Rise in Hb/protein > 5–6% during SBT, BNP rise, echo E/e′ ↑, lung US B-lines, PAOP > 18

Treatment

- Diuretics, nitrates, ischaemia treatment; extubate to NIV

Key point: Suspect it in COPD + heart disease who fails within minutes with hypertension.

4. ICU-acquired weakness and the diaphragm

ICU-AW

- Critical illness polyneuropathy/myopathy; MRC sum score < 48
- Risks: sepsis, MOF, hyperglycaemia, steroids, NMB, immobility
- Nerve conduction/EMG if atypical; exclude GBS, myasthenia, cord lesion

Diaphragm

- Ultrasound: thickening fraction < 20–30% or excursion < 1 cm = dysfunction
- Avoid over-assistance and over-sedation; inspiratory muscle training

Management

- Early mobilisation, glucose control, minimise steroids/NMB, nutrition, physiotherapy

Key point: Prevention beats treatment: light sedation and early mobilisation.

5. T-piece vs pressure support SBT

T-piece

- Closest to post-extubation work of breathing; unmasks cardiac failure
- Higher failure rate, may delay extubation

PSV 5–8

- Easier, compensates tube resistance (but upper airway oedema after extubation adds similar load)
- **Subirà 2019 (JAMA)**: 30 min PSV 8 → more successful extubation than 2 h T-piece

After extubation (high risk)

- NIV or HFNO+NIV (HIGH-WEAN) for > 65 yr, heart failure, COPD, obesity
- Cuff leak test in high-risk for stridor → steroids 12–24 h before

Key point: Choose PSV for most; T-piece when cardiac failure is suspected.

6. Tracheostomy

Benefits

- Less sedation, comfort, oral hygiene, ↓ dead space and resistance, easier weaning and mobilisation, ward discharge

Timing

- **TracMan 2013**: early (day 1–4) vs late (≥ day 10) → no mortality difference; 45% of late group never needed one
- Consider after ~10 days if weaning not imminent; earlier in neuro injury (SETPOINT2: no outcome benefit)

Technique and risks

- Percutaneous dilatational (bronchoscopy, US) vs surgical
- Bleeding, loss of airway, stenosis; NTSP safety bundles

Key point: Tracheostomy does not cut mortality; it facilitates weaning.

7. Prolonged weaning plan

Steps

- Identify and treat each cause; daily SBT or progressive reduction of PSV
- Tracheostomy mask trials (↑ duration), night rest on ventilator
- Specialised weaning unit, home NIV for hypercapnic COPD

Key point: Systematic: diagnose, correct, retest, rehabilitate.